

You said:

and communication passes through this central device. II. A _____ is set of rules that governs data communication.

ChatGPT said:

1. and communication passes through this central device.

👉 **Hub** (or **Switch**)

In a **star topology**, all communication passes through a **central device**, which can be a **hub** or a **switch**.

2. A _____ is set of rules that governs data communication.

👉 **Protocol**

A **protocol** is a set of rules that governs data communication between devices in a network.

Examples:

- HTTP
- FTP
- TCP/IP

If you want, I can give a quick 1-mark exam-ready definition also. 😊

You said:

b. Differentiate Hub, Switches and Routers. (02)

ChatGPT said:

b. Differentiate Hub, Switch and Router (02 Marks)

Basis	Hub	Switch	Router
Layer (OSI Model)	Physical Layer (Layer 1)	Data Link Layer (Layer 2)	Network Layer (Layer 3)
Data Handling	Broadcasts data to all devices	Sends data only to specific device (MAC address based)	Routes data between different networks (IP based)
Address Used	No address	MAC Address	IP Address
Speed & Efficiency	Slow, more collisions	Faster, fewer collisions	Fast, intelligent routing
Use	Connect devices in small LAN	Connect devices in LAN efficiently	Connect different networks (e.g., Internet)

Quick Exam Point (Very Short Answer)

- **Hub:** Simple device that sends data to all connected devices.
- **Switch:** Sends data only to the intended device using MAC address.
- **Router:** Connects different networks and uses IP address to route data.

If you want a 1-mark short differentiation version, I can give that too.

You said:

_____ is the process of adding a VLAN identification header to the frame. II. Which is a logical Partition of Layer 2 network? Answer in one word. b. What do you mean by root bridge in spanning tree protocol? (02) c. How does packet switching improve network efficiency?

ChatGPT said:

I. Fill in the blanks

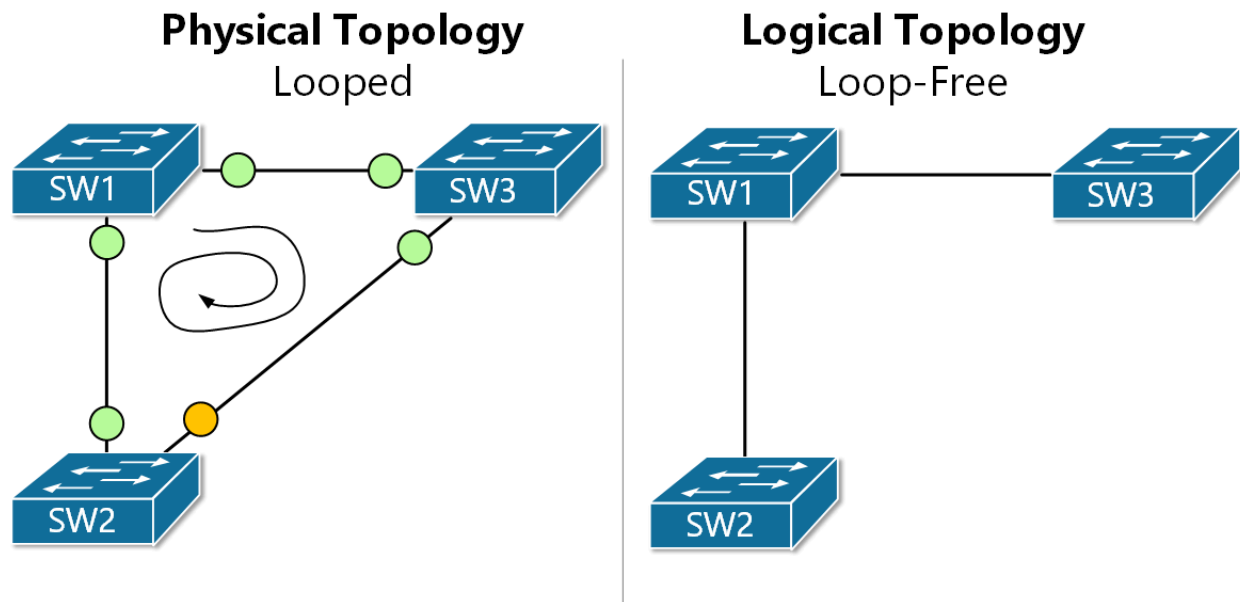
1) _____ is the process of adding a VLAN identification header to the frame.

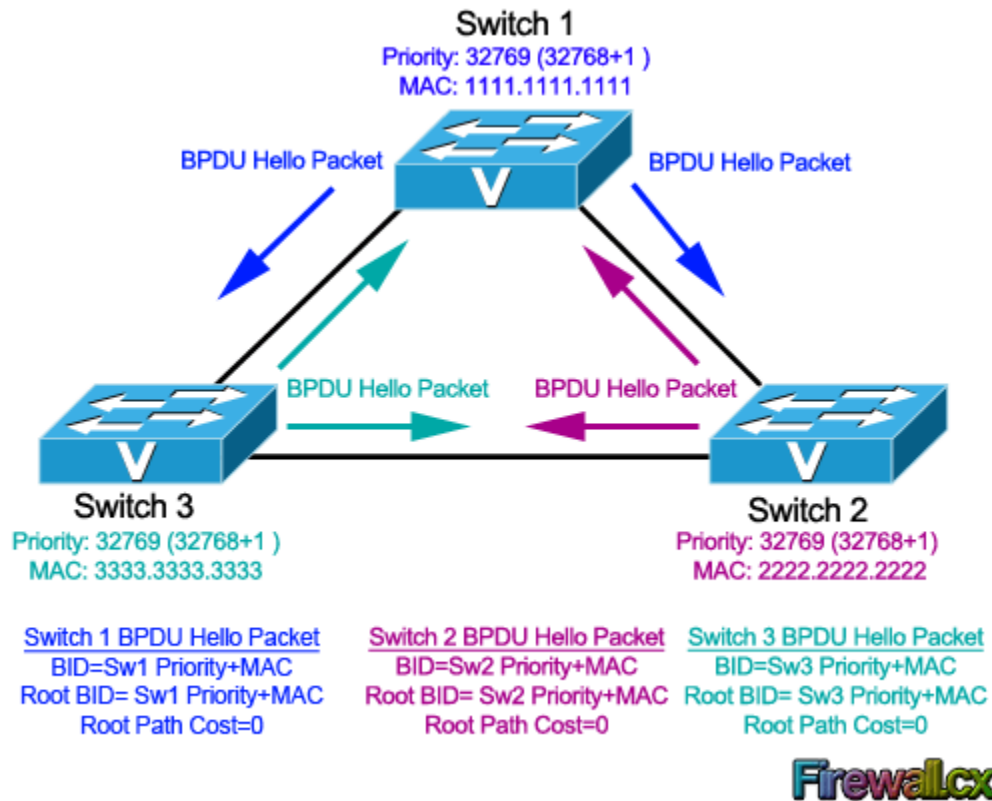
👉 Tagging (specifically VLAN Tagging / 802.1Q Tagging)

2) Which is a logical partition of Layer 2 network? (One word)

👉 VLAN

b. What do you mean by Root Bridge in Spanning Tree Protocol? (02 Marks)





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In **Spanning Tree Protocol (STP)**:

- The **Root Bridge** is the **main central switch** in the network.
- It is elected automatically based on the **lowest Bridge ID** (Priority + MAC Address).
- All other switches calculate the **shortest path to the Root Bridge**.
- It helps prevent **loops** in a Layer 2 network.

👉 In simple words:

Root Bridge is the reference switch in STP from which all path calculations are done.

c. How does Packet Switching improve network efficiency?

Packet switching improves efficiency by:

1. **Dividing data into small packets** instead of sending one large message.
2. **Sharing network bandwidth** among multiple users.

3. Sending packets through **different available paths**, reducing congestion.
4. Allowing better **utilization of network resources**.

👉 Result:

- Faster data transmission
- Less congestion
- More reliable communication

If you want very short 1–2 line exam answers, I can provide that too.

You said:

Two MCQs / Two fill in the blank questions. Each carry 1 marks. (02) I. Physical or logical arrangement of network is called _____. II. _____ layer is responsible for process-to-process delivery in a general network model b. Compare TCP/IP and OSI Reference Models. (02) c. Discuss Multiplexing with diagram. (04) Q.2 (08) a. Two MCQs / Two fill in the blank questions. Each carry 1 marks. (02) I _____ is the process of adding a VLAN identification header to the frame II. Which address is used by the Data Link Layer for communication? a) IP Address b) Port Number c) MAC Address d) Domain Name b. Discuss the function of RARP. (02) c. How does packet switching improve network efficiency?

ChatGPT said:

Q.1 (08)

a) Fill in the blanks (02)

I. Physical or logical arrangement of network is called

👉 **Topology**

II. _____ layer is responsible for process-to-process delivery in a general network model

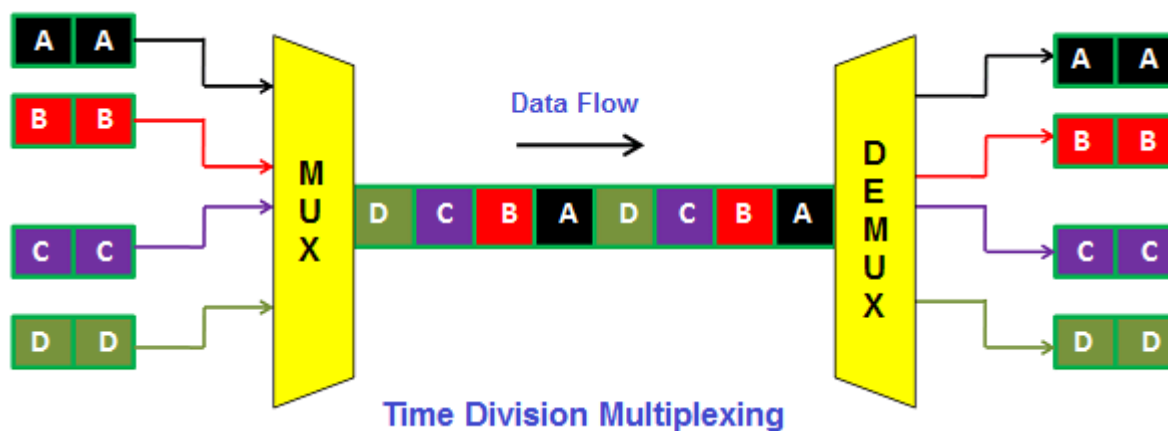
👉 **Transport Layer**

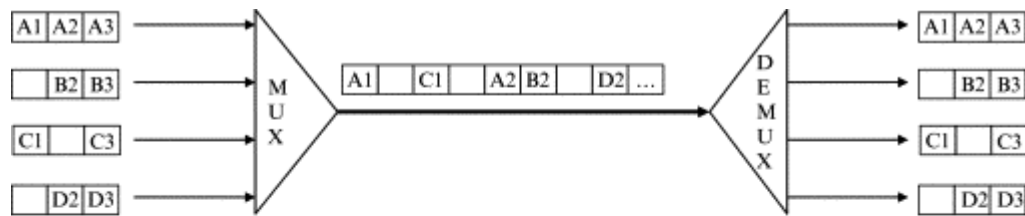
b) Compare TCP/IP and OSI Reference Models (02)

Basis	OSI Model	TCP/IP Model
Developed by	ISO	DARPA
Number of Layers	7 Layers	4 Layers
Approach	Theoretical model	Practical model
Layers Separation	Clear separation of services	Less strict separation
Example Layers	Physical to Application	Network Access to Application

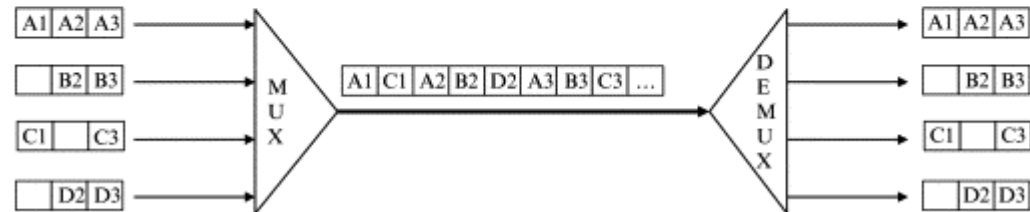
👉 **In short:** OSI is a conceptual reference model, while TCP/IP is a practical implementation used on the Internet.

c) Discuss Multiplexing with diagram (04)





(b)



(c)

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Multiplexing

Multiplexing is the technique of combining multiple signals into one single transmission medium.

Working:

1. Multiple input signals are given to a **Multiplexer (MUX)**.
2. The MUX combines them into one signal.
3. At the receiver side, a **Demultiplexer (DEMUX)** separates them.

Types:

- **FDM (Frequency Division Multiplexing)**
- **TDM (Time Division Multiplexing)**
- **WDM (Wavelength Division Multiplexing)**

👉 **Advantage:** Efficient use of bandwidth.

Q.2 (08)

a) Fill in the blanks / MCQ (02)

I. _____ is the process of adding a VLAN identification header to the frame

👉 **Tagging** (VLAN Tagging / 802.1Q Tagging)

II. Which address is used by the Data Link Layer for communication?

👉 Correct Answer: (c) MAC Address

b) Discuss the function of RARP (02)

RARP (Reverse Address Resolution Protocol) is used to obtain an **IP address** from a **known MAC address**.

Function:

- A device knows its **MAC address**.
- It sends a RARP request to the RARP server.
- The server replies with the corresponding **IP address**.

👉 Used mainly by diskless workstations (now replaced by DHCP).

c) How does packet switching improve network efficiency? (Repeated – 02/04 Marks)

Packet switching improves efficiency by:

1. Dividing data into smaller **packets**.
2. Allowing multiple users to share the same network.
3. Sending packets through the **best available path**.
4. Reducing congestion and improving bandwidth usage.

👉 Result: Better speed, reliability, and efficient resource utilization.

If you want ultra-short exam-ready answers (strict 2–3 lines per question), tell me.

You said:

Attempt following MCQ (02) I. Which of the following is the main responsibility of the Physical Layer? a) Error detection b) Routing c) Transmission of raw bits d) Encryption II. Which of the following is NOT a guided transmission medium? a) Twisted pair cable b) Fiber optic cable c) Coaxial cable d) Radio waves b. Discuss the difference between guided and unguided transmission media. (02) c. Explain OSI layer in detail. (04) Q.2 Attempt Following Questions.

(08) a. Attempt following MCQ (02) I What is used by the Data Link Layer to detect errors in frames? a) CRC b) Routing Table c) ARP Protocol d) Session Key II. Which address is used by the Data Link Layer for communication? a) IP Address b) Port Number c) MAC Address d) Domain Name b. Explain need of ARP & RARP

ChatGPT said:

Q.1 (08)

a) Attempt MCQ (02)

I. Which of the following is the main responsibility of the Physical Layer?

👉 **Correct Answer: (c) Transmission of raw bits**

II. Which of the following is NOT a guided transmission medium?

👉 **Correct Answer: (d) Radio waves**

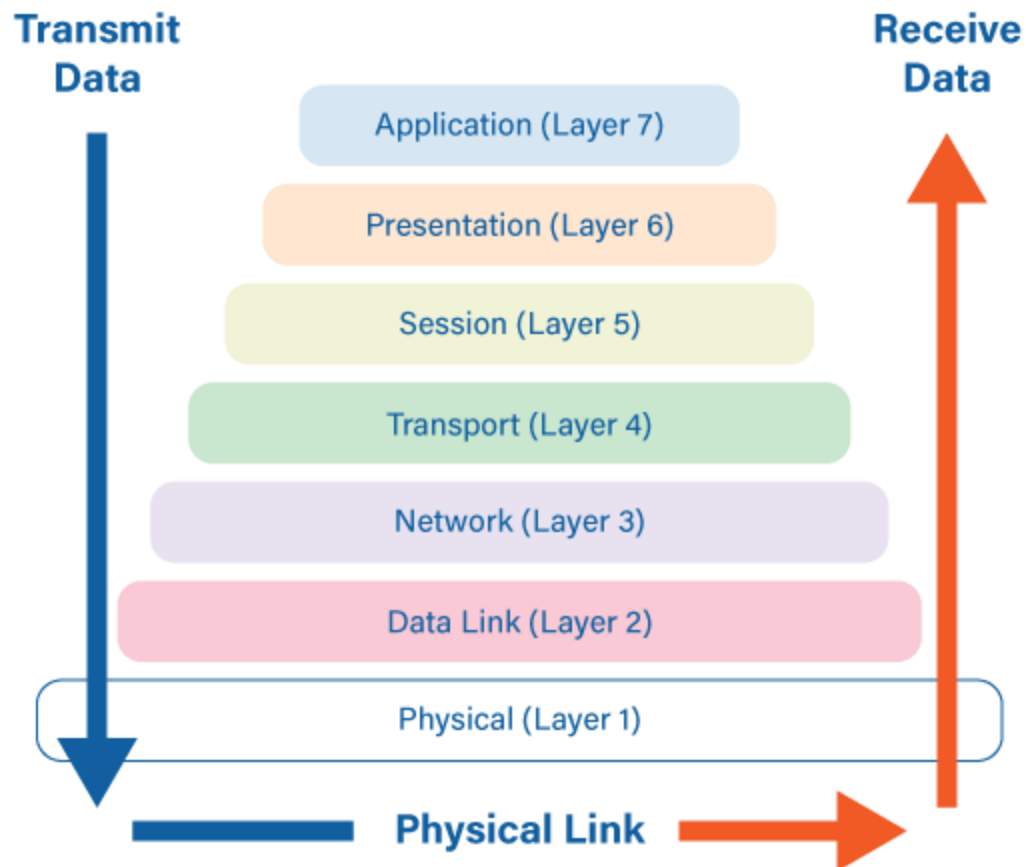
b) Difference between Guided and Unguided Transmission Media (02)

Basis	Guided Media	Unguided Media
Medium	Physical cable	No physical cable
Signal Path	Through wires	Through air/vacuum
Examples	Twisted pair, Coaxial, Fiber optic	Radio waves, Microwave, Satellite
Security	More secure	Less secure
Cost	Usually lower	Can be higher

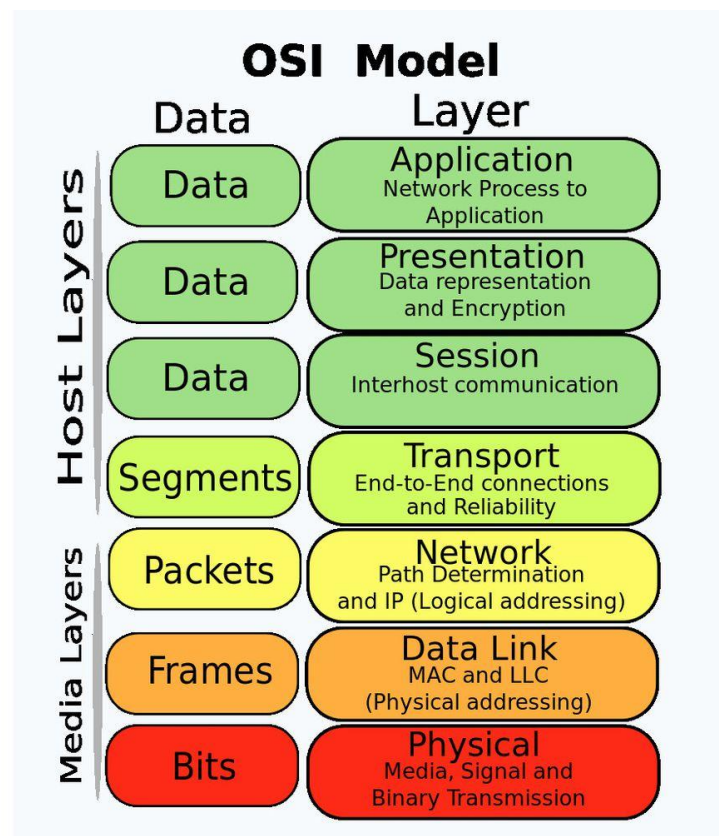
👉 Guided uses cables; Unguided uses wireless signals.

c) Explain OSI Model in detail (04)

The 7 Layers of OSI



Layer	Function	Example
Application (7)	Services that are used with end user applications	SMTP,
Presentation (6)	Formats the data so that it can be viewed by the user Encrypt and decrypt	JPG, GIF, HTTPS, SSL, TLS
Session (5)	Establishes/ends connections between two hosts	NetBIOS, PPTP
Transport (4)	Responsible for the transport protocol and error handling	TCP, UDP
Network (3)	Reads the IP address form the packet.	Routers, Layer 3 Switches
Data Link (2)	Reads the MAC address from the data packet	Switches
Physical (1)	Send data on to the physical wire.	Hubs, NICS, Cable



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The **OSI (Open Systems Interconnection) Model** has **7 layers**:

1 Physical Layer

- Transmits raw bits
- Defines cables, connectors, voltage levels

2 Data Link Layer

- Error detection (CRC)
- MAC addressing
- Frame formation

3 Network Layer

- Routing
- Logical addressing (IP)

4 Transport Layer

- End-to-end delivery
- Error control & flow control
- Protocols: TCP, UDP

5 Session Layer

- Establishes and manages sessions

6 Presentation Layer

- Encryption, Compression
- Data formatting

7 Application Layer

- Interface for users
- Protocols: HTTP, FTP, SMTP

👉 OSI model standardizes communication between different systems.

Q.2 (08)

a) Attempt MCQ (02)

I. What is used by the Data Link Layer to detect errors in frames?

👉 Correct Answer: (a) CRC

II. Which address is used by the Data Link Layer for communication?

👉 Correct Answer: (c) MAC Address

b) Explain need of ARP & RARP (04)

ARP (Address Resolution Protocol)

- Used to find **MAC address from known IP address**.
- Required for communication within a local network.

👉 Example: If a device knows the IP but not the MAC, ARP helps find it.

RARP (Reverse Address Resolution Protocol)

- Used to find **IP address from known MAC address**.
- Mainly used by diskless workstations.
- Now mostly replaced by DHCP.

👉 ARP: IP → MAC

👉 RARP: MAC → IP

If you want strict 2-mark short answers only, I can compress this further.

You said:

. _____ layer provides the services to user. II. Physical or logical arrangement of network is called _____. b. You have been asked to build a 10 Gbps fiber optic connection over a 5 km link using CSMA/CD as the channel access protocol. What is the minimum frame size required? (02) c. Find the encoded data which is sent by sender: The data bit to be sent is

[100100], and the divisor is [1101] (04) Q.2 (08) a. Two MCQs / Two fill in the blank questions. Each carry 1 marks. (02) I _____ address is used to identify a process on a host by the transport layer. II. _____ is the best example of an Access Point of wireless network . b. Explain even and odd parity.

ChatGPT said:

Q.1 (08)

a) Fill in the blanks (02)

I. _____ layer provides the services to user.

👉 Application Layer

II. Physical or logical arrangement of network is called _____

👉 Topology

b) Minimum Frame Size Calculation (02)

Given:

- Bandwidth = **10 Gbps** = 10^{10} bits/sec
- Distance = **5 km**
- Assume propagation speed in fiber $\approx 2 \times 10^8$ m/s

Step 1: Propagation Delay

$$\begin{aligned}\text{Propagation Delay} &= \frac{5000}{2 \times 10^8} \\ &= 25 \text{ microseconds}\end{aligned}$$

Step 2: Round Trip Time (RTT)

$$RTT = 2 \times 25 = 50 \text{ microseconds}$$

Step 3: Minimum Frame Size

$$\begin{aligned}\text{Minimum Frame Size} &= \text{Bandwidth} \times RTT \\ &= 10^{10} \times 50 \times 10^{-6}\end{aligned}$$

= 500,000 bits

👉 **Minimum Frame Size = 500,000 bits ($\approx$ 62,500 bytes)**

c) CRC Encoding (04)

Data = **100100**

Divisor = **1101**

Step 1: Append 3 zeros (since divisor length = 4)

Data becomes:

100100000

Step 2: Perform Binary Division (Modulo-2)

After division, the **remainder = 001**

Step 3: Encoded Data (Data + Remainder)

100100001

👉 **Encoded data sent = 100100001**

Q.2 (08)

a) Fill in the blanks (02)

I. _____ address is used to identify a process on a host by the transport layer.

👉 **Port Address (Port Number)**

II. _____ is the best example of an Access Point of wireless network.

👉 **Wi-Fi Router**

b) Explain Even and Odd Parity (04)

Parity Bit

A **parity bit** is used for simple error detection in data communication.

1 Even Parity

- Total number of 1s (including parity bit) is **even**.
- If data has odd number of 1s, parity bit = 1.
- If data has even number of 1s, parity bit = 0.

Example:

Data = 1011 (3 ones → odd)

Parity bit = 1

Transmitted = 10111 (4 ones → even)

2 Odd Parity

- Total number of 1s (including parity bit) is **odd**.
- If data has even number of 1s, parity bit = 1.
- If data has odd number of 1s, parity bit = 0.

Example:

Data = 1011 (3 ones → already odd)

Parity bit = 0

Transmitted = 10110

👉 Parity detects **single-bit errors**, but cannot correct errors.

If you want step-by-step CRC long division shown fully, I can write it in detailed format.

You said:

solve full paper

ChatGPT said:

✅ Q.1

a) One Mark Questions

a. The data in the Physical Layer is represented in which form?

👉 **Bits (Electrical/Optical signals – voltage/energy level)**

b. The measure of how much data can be transmitted in a given time period is called

👉 **Bandwidth (Data Rate)**

c. A device that regenerates and amplifies signals in the Physical Layer is called

👉 **Repeater**

d. _____ multiplexing technique is different signals transmitted at different time slots?

👉 **TDM (Time Division Multiplexing)**

e. _____ Multiple Access Protocol primarily used in wireless network.

👉 **CSMA/CA**

✅ Q.2 (05)

a) Final encoded 7-bit even parity Hamming code for data bits 1010 (03)

For Hamming (7,4):

Parity bit positions → 1, 2, 4

Data bit positions → 3, 5, 6, 7

Place data 1010:

Position	1	2	3	4	5	6	7
Bit	P1	P2	1	P4	0	1	0

Now calculate even parity:

P1 covers: 1,3,5,7 → (P1,1,0,0) → total 1 → P1 = 1

P2 covers: 2,3,6,7 → (P2,1,1,0) → total 2 → P2 = 0

P4 covers: 4,5,6,7 → (P4,0,1,0) → total 1 → P4 = 1

Final Code:

👉 **1011010**

b) Transmission Impairment (02)

Transmission impairment refers to the **degradation of signal strength or quality during transmission.**

Types:

1. **Attenuation** – Signal becomes weak
 2. **Distortion** – Signal shape changes
 3. **Noise** – Unwanted signals interfere
-

OR

a) Propagation Delay Calculation (02)

Distance = 1.5 km

Speed = 225,000 km/sec

Propagation Delay

$$= \frac{1.5}{225000}$$

$$= 6.67 \times 10^{-6} \text{ s}$$

$$\text{Delay} = \frac{\text{Distance}}{\text{Speed}}$$

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👉 **Propagation delay = 6.67 microseconds**

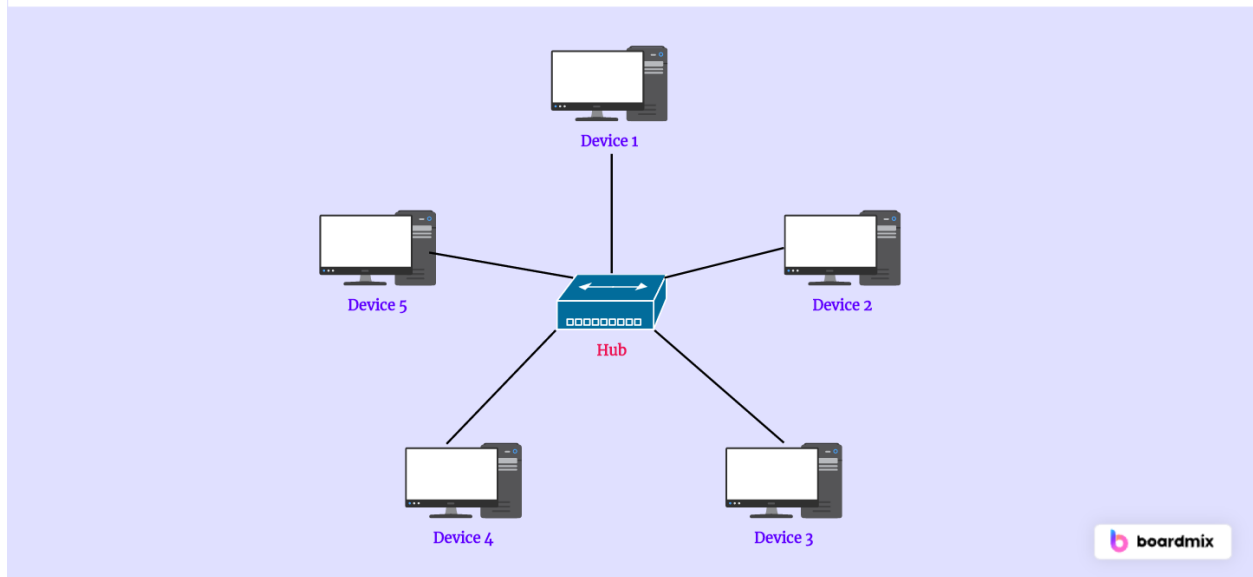
b) What is Topology? Explain any two types (03)

Topology

Topology is the **physical or logical arrangement of devices in a network.**

1 Star Topology

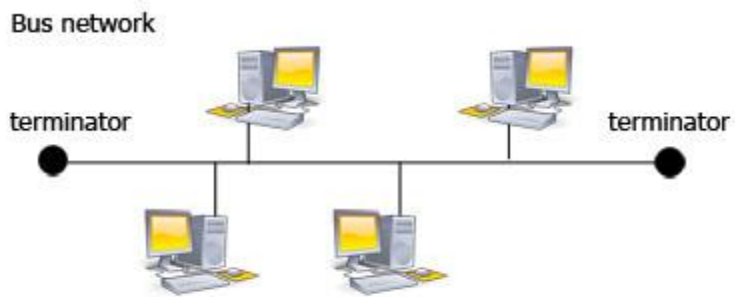
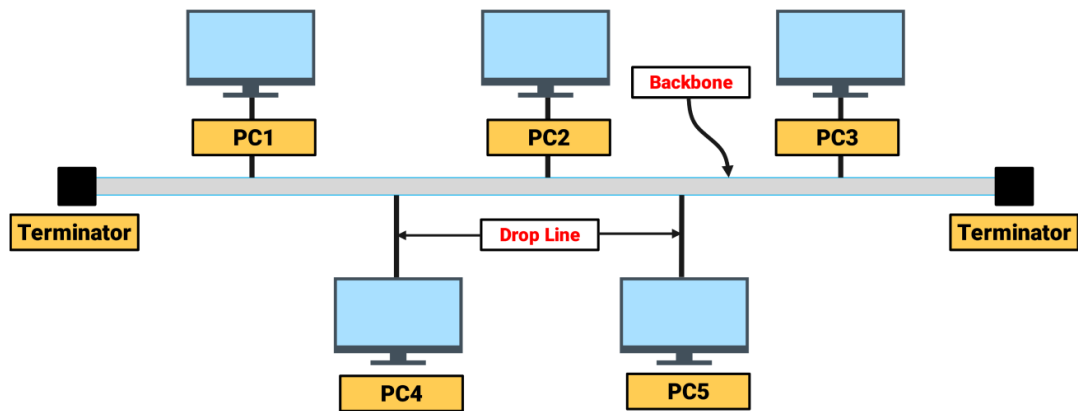
Star Topology Diagram



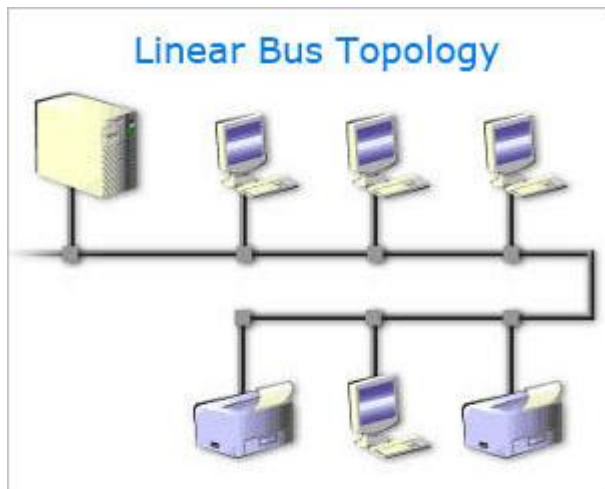
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- All devices connected to central hub/switch
- Easy to manage
- Failure of central device stops network

2 Bus Topology



(c) teach-ict.com



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- Single main cable
 - All devices share medium
 - Cable failure affects entire network
-

✅ Q.3 (05)

Explain Cyclic Redundancy Check (CRC)

Given:

Data = **100100**

Divisor = **1101**

Step 1: Append 3 zeros

100100000

Step 2: Perform Modulo-2 division

Remainder = **001**

Step 3: Final transmitted data

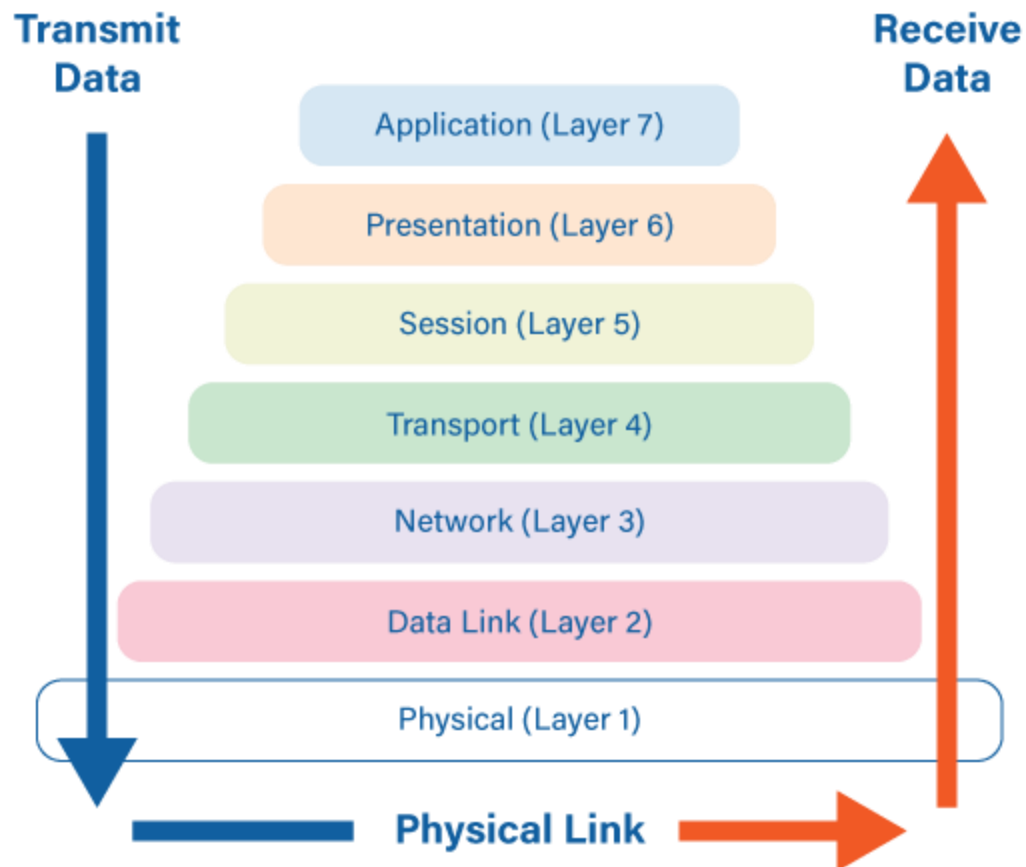
👉 **100100001**

CRC is used for **error detection** in Data Link Layer.

✅ Q.4 (05)

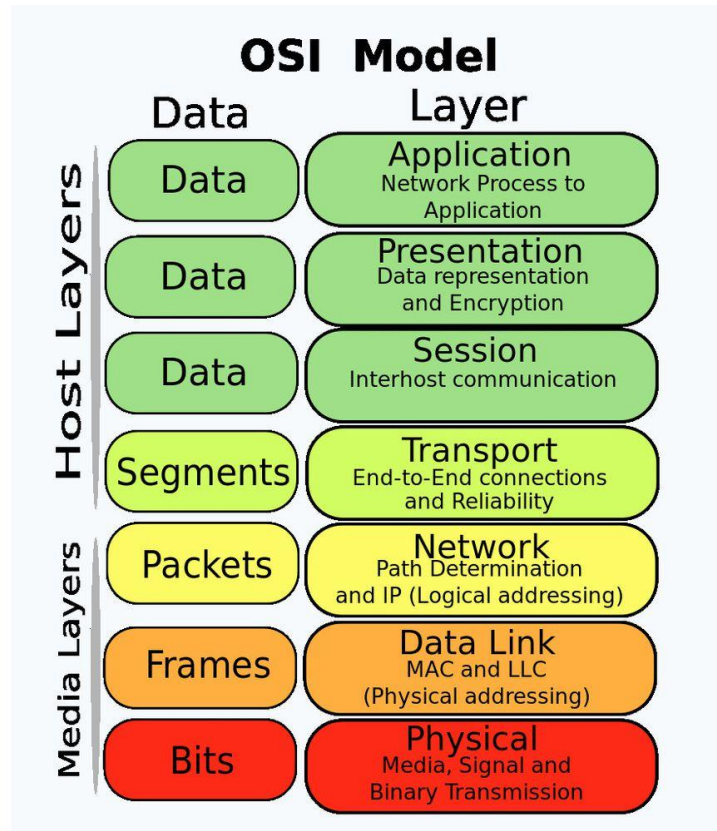
OSI Layers with functionality

The 7 Layers of OSI



OSI (Open Source Interconnection) 7 Layer Model

Layer	Application/Example	Central Device/ Protocols		DOD4 Model
Application (7) Serves as the window for users and application processes to access the network services.	End User layer Program that opens what was sent or creates what is to be sent Resource sharing • Remote file access • Remote printer access • Directory services • Network management	User Applications SMTP	G A T E W A Y Can be used on all layers	Process
Presentation (6) Formats the data to be presented to the Application layer. It can be viewed as the "Translator" for the network.	Syntax layer encrypt & decrypt (if needed) Character code translation • Data conversion • Data compression • Data encryption • Character Set Translation	JPEG/ASCII EBDIC/TIFF/GIF PICT		
Session (5) Allows session establishment between processes running on different stations.	Synch & send to ports (logical ports) Session establishment, maintenance and termination • Session support - perform security, name recognition, logging, etc.	Logical Ports RPC/SQL/NFS NetBIOS names		
Transport (4) Ensures that messages are delivered error-free, in sequence, and with no losses or duplications.	TCP Host to Host, Flow Control Message segmentation • Message acknowledgement • Message traffic control • Session multiplexing	F I L T E R I N G P A C K E T	TCP/SPX/UDP	Host to Host
Network (3) Controls the operations of the subnet, deciding which physical path the data takes.	Packets ("letter", contains IP address) Routing • Subnet traffic control • Frame fragmentation • Logical-physical address mapping • Subnet usage accounting		Routers IP/IPX/ICMP	Internet
Data Link (2) Provides error-free transfer of data frames from one node to another over the Physical layer.	Frames ("envelopes", contains MAC address) [NIC card — Switch — NIC card] (end to end) Establishes & terminates the logical link between nodes • Frame traffic control • Frame sequencing • Frame acknowledgment • Frame delimiting • Frame error checking • Media access control	Switch Bridge WAP PPP/SLIP	Land Based Layers	Network
Physical (1) Concerned with the transmission and reception of the unstructured raw bit stream over the physical medium.	Physical structure Cables, hubs, etc. Data Encoding • Physical medium attachment • Transmission technique - Baseband or Broadband • Physical medium transmission Bits & Volts	Hub		



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1 Physical Layer

- Transmits raw bits
- Defines cables & voltage levels

2 Data Link Layer

- Framing
- MAC addressing
- Error detection (CRC)

3 Network Layer

- Routing
- Logical addressing (IP)

4 Transport Layer

- End-to-end delivery

- Flow & error control

5 Session Layer

- Session establishment & management

6 Presentation Layer

- Encryption
- Compression
- Format translation

7 Application Layer

- Provides services to users
- HTTP, FTP, SMTP

 **✓ Full Paper Completed**

If you want this

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